JUNXIANG YANG

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EDUCATION

Nanyang Technological University *Jan. 2025 – Present*
Ph.D. in Electrical and Electronic Engineering GPA: 4.17/5.0

University of Science and Technology of China *Sept. 2020 – June 2024*
B.S. in Physics Major: Applied Physics GPA: 3.21/4.3
Selected Courses: Electronic Circuits (93), Principle of Microcomputer (90)

Nanyang Technological University *July 2023 – Oct. 2023, Feb. 2024 – Apr. 2024*
Research internship in School of Electrical and Electronic Engineering

EXPERIENCES

Research Assistant / Ph.D. Student *Sept. 2024 – Present*
Advisor: Prof. Yun Yang *NTU EEE, Singapore*

- Designed **PCB-based electromagnetic metasurfaces** for MHz wireless power transfer and validated their voltage enhancement through simulation and experiments.
- Developed **two-stage FEM-assisted optimization strategies** for compact PCB-based resonant couplers in capacitive and hybrid wireless power transfer systems.
- Optimized **sandwich-structured PCB resonators** to improve equivalent quality factor, system efficiency, and misalignment tolerance.
- Built **data-driven prediction models** for equivalent-circuit electrical parameters of PCB-based couplers to accelerate design iteration.

Development of Host Software for X-ray Photoelectron Spectroscopy *May 2024 – Apr. 2025*
Advisor: Prof. Changqing Feng *USTC, Hefei*

- Designed a **user interface** using PyQt5, providing an intuitive GUI for experimental operation.
- Developed and tested a communication module for reliable information exchange with FPGA and detector.

PCB-Based Wireless Power Resonator Research Internship *July 2023 – Oct. 2023, Feb. 2024 – May 2024*
Advisor: Prof. Yun Yang *NTU, Singapore*

- Simulated polygonal planar spiral coils in HFSS and validated the design with PCB prototypes.
- Designed and optimized **13.56 MHz PCB resonators** with high quality factor for wireless power transfer.
- Developed an automated optimization workflow combining machine learning and finite element simulation.

Design of High-Voltage Circuits for MeV Gamma-Ray Detector *July 2022 – May 2023*
Advisor: Prof. Changqing Feng *USTC, Hefei*

- Designed the schematic and PCB of the power module for a MeV gamma-ray detector.
- Validated the power module and mitigated noise issues.

TECHNICAL STRENGTHS

Computer Languages	Python, MATLAB
Simulation Software	Ansys HFSS, Ansys Maxwell, COMSOL Multiphysics
English	TOEFL (87), IELTS (7.5), GRE (319+3.5)

HONORS AND AWARDS

Best Paper Award, 20th IEEE Conference on Industrial Electronics & Applications (ICIEA, Aug. 2025)
Finalist Certificate, 2024 IEEE Global Student Wireless Power Competition (SWPC, May 2024)
Certificate of Merit & **Certificate of Third Prize**, The Student Competition on Electric Transportation Design – Automotive, Vessel and Aircraft (May 2024)

LEADERSHIP

President of the Student Union, School of Physical Science *Apr. 2022 – Apr. 2023*

PUBLICATIONS

- **Junxiang Yang**, Xiangrong Zhang, Yici Wang, Shuye Shang, Kaiyuan Wang, Yun Yang, "A Frequency-Adjustable PCB Shielding Coil in Wireless Power Transfer System," in *Wireless Power Transfer*, Jul. 2024.
- **Junxiang Yang**, Kaiyuan Wang, Jiayang Wu, Yun Yang, "Two-Stage Optimization of DC Equivalent Series Resistances in Compact PCB-Based Resonant Couplers for Capacitive Power Transfer," in *20th IEEE Conference on Industrial Electronics and Applications (ICIEA)*, Yantai, China, Aug. 2025, pp. 1–6.
- **Junxiang Yang**, Kaiyuan Wang, Zhen Sun, Junming Zeng, "A Two-Stage Optimization Strategy for Enhancing the Quality Factors of Sandwich-Structured PCB Based Resonators," in *2025 IEEE Transportation Electrification Conference and Expo, Asia-Pacific (ITEC Asia-Pacific)*, Singapore, Nov. 2025.
- Kaiyuan Wang, **Junxiang Yang**, Zhen Sun, Yun Yang, "Optimized Stacked Passive Coil Array Design for High Efficiency and Misalignment Tolerance in PCB-Based Wireless Power Transfer Systems," in *IEEE Transactions on Power Electronics*, 2026.
- Yao Wang, **Junxiang Yang**, Y. Song, Kaiyuan Wang, Yun Yang, "Three-Mode MHz Wireless Power Transfer: Inductive, Capacitive, and Hybrid, with a Unified Coupling Architecture," in *IEEE Journal of Wireless Power Technologies*, 2026.
- Yao Wang, **Junxiang Yang**, Kaiyuan Wang, Yun Yang, "Highly Integrated Hybrid Inductive and Capacitive Power Transfer System with Asymmetrical Printed-Circuit-Board-Based Self-Resonator," in *IEEE Transactions on Power Electronics*, vol. 40, no. 7, pp. 10254–10264, July 2025.
- Kaiyuan Wang, **Junxiang Yang**, Zhen Sun, Junming Zeng, and Yun Yang, "Analysis and design of domino inductive power transfer system with enhanced quasi-load-independent constant current output," in *2024 IEEE Wireless Power Technology Conference and Exposition (WPTCE)*, May 2024.
- Yajie Jiang, **Junxiang Yang**, Jiayang Wu, Kerui Li, Yun Yang and S. C. Tan, "A secured fast charging control based on a machine learning-aided electrothermal model for lithium-ion batteries," in *2024 IEEE Energy Conversion Congress and Exposition (ECCE)*, Oct. 2024.
- Shuye Shang, **Junxiang Yang**, Tenghao Ji, Minghao Fan, Kaiyuan Wang, and Heshou Wang, "Investigations of a Simplified PCB-Based Wireless Power Resonator Operating at 13.56 MHz," in *10th International Conference on Power Electronics Systems and Applications (PESA)*, Jun. 2024.